Lecture plan · for the lecturer

Notional machines and tracing

Programming · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Define notional machines.
- Recognise misconceptions.
- Design diagnostic questions.

Activity: Write diagnostic questions

1. Warm-up (5 min): Show the three-line trace and ask everyone to vote before discussing.
2. Teach (10 min): Present five misconceptions from Sorva's catalogue with examples.
3. Write (25 min): Pairs write one tracing question whose distractors each match a misconception.
4. Swap (15 min): Pairs swap questions and check which model each wrong answer reveals.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which misconception surprised you?
- What makes a distractor diagnostic?
- Where will you draw memory?

Exit ticket

- Write one distractor and the misconception it detects.

Try it in your next lecture

Start your next lecture with three tracing questions whose distractors each match a known misconception; students vote on their phones, then redraw memory for the most-missed question.

Seminar sheet · for students

Notional machines and tracing

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Name _____ Group _____ Date _____

1 **A student answers 7 7. Which misconception does that reveal?**

◯ Print shows the last line only ◯ Assignment creates a lasting link ◯ Variables can't change

2 **Trace the swap.**

```
x = 1
y = 2
t = x
x = y
y = t
```

After this code runs, x is _____ and y is _____.

Word bank: 3 · 2 · 1 · 0

3 **Which are documented novice misconceptions? Choose all that apply.**

☐ Each call has its own local variables ☐ A variable can hold several values at once

☐ The computer infers what the programmer meant ☐ Assignment links variables permanently

4 **A student can write correct code on a quiz without having a viable model of assignment.**

◯ True ◯ False

5 **Match each wrong answer to the misconception it reveals.**

b = 5; a = b; b = 7 → a is 7	• • • •	Definition is execution
x = 3; x = 4 → x is 3 and 4	• • • •	Assignment as a link
A defined function runs without being called	• • • •	The computer knows intent
Misspelt name still works	• • • •	A variable holds history

Answers · for the lecturer

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1 **Assignment creates a lasting link**

That is the link model of assignment.

2 **After this code runs, x is 2 and y is 1.**

The temporary t keeps the old x, so the values swap; without t, both would end as 2.

3 **Assignment links variables permanently; A variable can hold several values at once; The computer infers what the programmer meant**

Separate locals per call is the correct model, not a misconception.

4 **True**

Quiz code can pass by pattern-matching; tracing questions reveal the model.

5 **b = 5; a = b; b = 7 → a is 7 — Assignment as a link**
x = 3; x = 4 → x is 3 and 4 — A variable holds history
A defined function runs without being called — Definition is execution
Misspelt name still works — The computer knows intent

Each distractor should point to exactly one model.

For you to think about

Which misconception do you suspect is common in your course but have never measured?